Not In Our Time: Transport-Based Content Selection as an Inspectable, Manipulable Alternative to RAG

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Retrieval-Augmented Generation [Lewis et al., 2020] as typically deployed [Gao et al., 2024]—including in Google's NotebookLM podcast generator [Google, 2024]—retrieves by embedding similarity and feeds results to an LLM. This is effective but opaque: the user cannot see *why* a passage was selected, cannot adjust selection criteria, and cannot explore how different editorial priorities reshape the output. We present an alternative in which passage selection is a min-cost flow problem [Ahuja et al., 1993, Peyré and Cuturi, 2019], making every selection decision visible as a cost, a flow, and a constraint. Because solving a flow is cheap compared to regenerating with an LLM, the system gives its user a flexible thinking tool for rapid exploration of alternatives. We demonstrate it by generating imitation panel discussions in the style of BBC Radio 4's *In Our Time* [Bragg, 1998–present], focussing on Charles Dickens' legally-themed *Bleak House* [Dickens, 1852–1853].

The system operates over 6,916 passages enriched by an LLM with structured metadata and contextual embeddings [Anthropic, 2024] (20 fields covering narrative function, thematic weight, and emotional register). The first transport problem selects passages to satisfy *demand profiles* declared by fictional expert personas—a novelist, a legal historian, a close reader, a performer, a Marxist, a traditionalist. The second assigns them to episode segments. An LLM generates multi-voice dialogue with speaking-style annotations for TTS rendering.

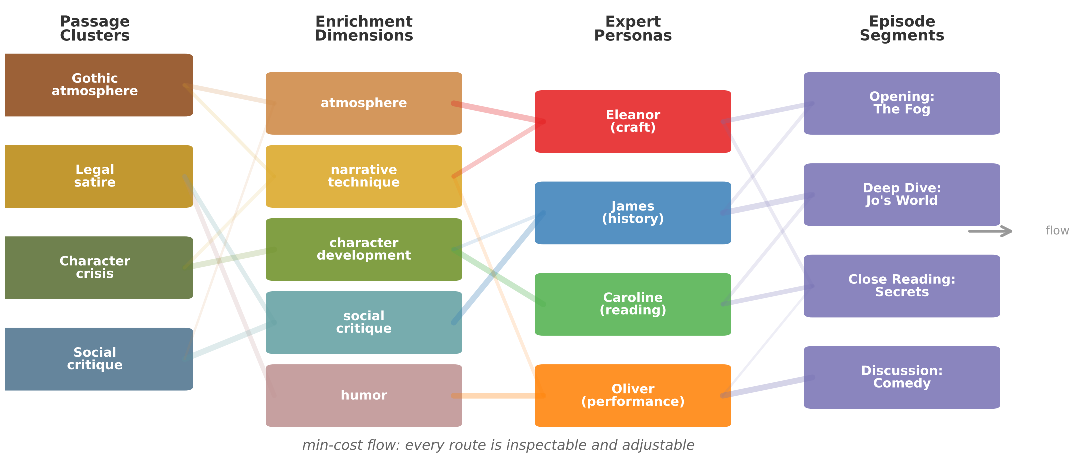

Figure 1: Passages flow through enrichment dimensions and expert demand nodes to episode segments. Every arc is inspectable and adjustable.

The expert personas are, deliberately, *stereotypes*: the Marxist always finds class struggle; the performer always finds comedy; the close reader always finds prose rhythm. The generated discussions are closer to parody than scholarship—each configuration produces a recognisably different *caricature* of academic discourse. This is the point. By making the stereotypes explicit and adjustable—encoded as demand vectors rather than implicit in a prompt—the system turns editorial bias into a first-class object that can be inspected, compared, and argued about. Every editorial decision is inspectable and manipulable. Twenty configurations, varying panel composition and arc emphasis, produce measurably different outputs: different passage selections (mean pairwise Jaccard similarity 0.52), different vocabulary, and different grammatical registers, confirmed by corpus comparison methods [Kilgarriff, 2001]. What happens when we remove the social historian? Double a character arc? Replace the close reader with a comedian? The answers are computable, and the differences are legible.

We do not claim the generated podcasts are good—they are fluent, structured, and plausible-sounding, which is precisely the problem with LLM-generated academic discussion [Doshi and Hauser, 2024]. Following Boden's [2004] taxonomy, the system performs combinational creativity, but the *interesting* creativity is the user's: exploring the configuration space and confronting the stereotypes that make the machinery work. The system's value is diagnostic, turning a black-box pipeline into a navigable space of configurations. It is no accident that *Bleak House* centres on the law: lawyers reviewing documents under constraints need inspectable, manipulable thinking tools at least as much as literary critics do.

Keywords: optimal transport, min-cost flow, RAG, LLM stereotypes, editorial decision spaces

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